



Log-Location-Scale Survival Models for Heart Failure Data: A Comparative Study

By:

Jood Basem Alblaji - 2312190

Meral Hashim Aljohani - 2317271

Supervised by:

Dr. Dawlah Alsulami

Faculty of Sciences - Department of Statistics
King Abdulaziz University
Jeddah, Saudi Arabia

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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

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Abstract

Heart Failure (HF) is a major cause of death worldwide , characterized by highly variable survival patterns. In this study, parametric survival analyses within the Log-Location Scale (LLS) framework were applied to a clinical dataset of HF records to explain time-to-event dynamics and identify key clinical predictors. An automated backward elimination procedure optimized by the Akaike Information Criterion (AIC) narrowed the initial clinical features down to a smaller set of generalized variables : age, ejection fraction, serum creatinine, serum sodium, sex, and blood pressure. Model comparison across information criteria benchmarks identified the Weibull distribution as the optimal framework. A final significance layer pruned non-significant covariates, leaving age , ejection fraction , and serum creatinine as the key variables driving patient prognosis. Advanced age and elevated serum creatinine reduced survival time , while higher ejection fraction showed a protective effect. The final model fit was validated using Kaplan-Meier empirical alignment and fitted cumulative hazard , indicating a good overall fit.

Keywords: Accelerated Failure Time, Heart Failure, Log-Location Scale, Weibull Distribution, Parametric Modeling.

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List of Symbols

Symbol	Description
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
CAIC	Consistent Akaike Information Criterion
VIF	Variance Inflation Factors
LLS	Log-Location-Scale
AFT	Accelerated Failure Time
HF	Heart Failure
SD	Standard Deviation
MLE	Maximum Likelihood Estimation
ML	Maximum Likelihood
RF	Random Forests
logL	Log-Likelihood
SEV	Smallest Extreme Value
PDF	Probability Density Function
CDF	Cumulative Density Function

Chapter 1: Introduction

Chapter 1

Introduction

1.1 Introduction

Heart Failure (HF) is a chronic and progressive clinical syndrome. It occurs when the heart muscle is structurally or functionally impaired, making it unable to pump enough blood to meet the body's metabolic demands for oxygen and nutrients. Medically, this inadequate cardiac output leads to fluid accumulation (congestion) in the lungs and peripheral extremities. The physiological danger of HF to the human body is severe; if left unmanaged, it heavily restricts physical mobility, causes secondary organ damage, and significantly elevates the risk of sudden cardiac death. Globally, HF is a critical public health burden affecting over 26 million individuals (Savarese and Lund, 2017). The annual mortality rates and overall survival probabilities are heavily dictated by demographic factors, particularly gender and age (Savarese and Lund, 2017). Epidemiological statistics demonstrate that males generally exhibit a higher incidence rate and poorer long-term survival outcomes compared to females. Furthermore, the mortality rate accelerates drastically among the elderly population. While the general 5-year survival rate for HF patients hovers around 50% to 57%, the prognosis worsens steeply with advancing age, with patients over the age of 75 experiencing significantly truncated survival timelines and reduced life expectancy after diagnosis (Jones et al., 2019). However, its occurrence in the Kingdom of Saudi Arabia poses particular clinical and epidemiological difficulties. Patients with HF in Saudi Arabia are diagnosed much earlier than those in the West, usually ten years earlier (Elasfar et al., 2020). A high prevalence of metabolic comorbidities,

especially ischemic heart disease, severe type 2 diabetes mellitus, and chronic hypertension, significantly accelerate the national burden of HF. Due to high re-hospitalization rates and unique biochemical risk profiles, recent domestic registries demonstrate that the clinical presentation of HF in Saudi hospitals necessitates specialized tracking. The prevalence of primary metabolic comorbidities and baseline clinical features among HF admissions in Saudi Arabian hospitals is presented in Figure 1.1.

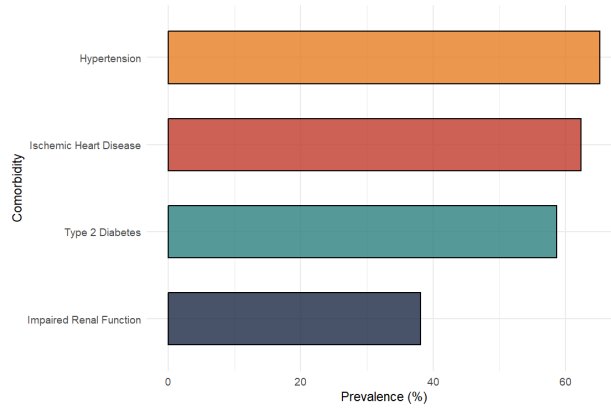


Figure 1.1: Prevalence of Comorbidities in Saudi HF Patients

As shown in Figure 1.1, a very complex and early-onset clinical profile is highlighted by the high incidence of type 2 diabetes 58.7%, ischemic heart disease 62.4%, and chronic hypertension 65.2%. In order to accurately capture local survival dynamics, specialized statistical modeling is required due to the accelerated disease progression caused by these intersecting physiological risk factors.

1.2 Research Problem

Despite substantial advancements in cardiovascular care, HF continues to be one of the foremost causes of mortality globally. The key challenge associated with HF data is the high level of complexity and heterogeneity. The distributional properties of such data cannot be sufficiently reflected by

a single model, as the skewed and heterogeneous nature of survival times means that no single model captures the data in the same way as others. Therefore, using a single modeling technique can result in an incomplete or even misleading representation of the underlying survival behavior.

For the analysis of survival data, there are different parametric regression models. Each of these models is based on a different distributional assumption, and each model interprets covariate effects differently. The models differ significantly in their fit to the data and in the description of its main features.

Given these considerations, the choice of an appropriate model is an important issue in the analysis of HF survival data. As a result, it is necessary to conduct a systematic comparison of competing models in order to determine which best explains the observed data. The present study is concerned with fitting and comparing a number of parametric regression models to determine the best model to describe the survival times of HF.

1.3 Research Importance

This study is important as it seeks to improve the modeling of HF survival data. The time to event data are complex and heterogeneous which require establishing an appropriate statistical model to adequately describe the underlying structure. Different parametric models make different distributional assumptions that can lead to different model fit and interpretation of the covariate effects.

The current study stresses the importance of comparing different modeling approaches using several parametric regression models rather than a single specification. Comparing models such as the Weibull, log-normal and log-logistic distributions when applied to real HF data can provide useful insight into the relative performance of the models.

Gaining a better understanding of the variables that speed up or slow down disease progression can help in optimizing the treatment. Statistically, this highlights the need to choose the model that provides a good fit to the data. Therefore, the model offers meaningful interpretations of the effect of

covariates and ensures that the inferences made from the analysis are valid and insightful.

1.4 Research Objectives

In this study, the primary focus is on statistically modeling the survival times of HF patients by using parametric survival distributions within the Accelerated Failure Time (AFT) framework. The specific objectives are as follows:

- Fit three parametric survival models from the Log-Location-Scale (LLS) family namely, the Weibull, log-normal, and log-logistic distributions to the HF dataset.
- Identify and select the most significant covariates in the HF dataset using an appropriate variable selection procedure.
- Evaluate and compare the goodness-of-fit of these competing models using some information criteria, in order to determine the model that best fits the data.
- Identify and quantify the effect of key covariates in the HF dataset on survival time using the selected model.

1.5 Definitions and Concepts

To ensure clarity, Table 1.1 presents key concepts and terms used throughout the study.

Table 1.1: Definition of Study Concepts and Statistical Models

Concept	Detailed Definition and Context
Heart Failure	A progressive clinical condition in which the heart is unable to pump enough blood to meet the body's needs, leading to reduced organ perfusion and overall physiological imbalance.
Survival Analysis	A branch of statistics concerned with analyzing the time until the occurrence of an event of interest, such as death or system failure.
Probability Density Function	A function that describes how the values of a continuous random variable are distributed. It provides the relative likelihood of observing a specific value of the variable.
Cumulative Distribution Function	A function that represents the probability that a random variable takes a value less than or equal to a given point, defined as $F(t) = P(T \leq t)$.
Survival Function	A function that gives the probability that the survival time exceeds a specified time t , defined as $S(t) = P(T > t) = 1 - F(t)$.
Hazard Function	A function that describes the instantaneous rate at which the event occurs at time t , given that it has not occurred before that time. It is defined as $h(t) = \frac{f(t)}{S(t)}$.
Statistical Models	
AFT Model	A fully parametric model that measures the effect of covariates on survival time, interpreting them as acceleration/deceleration factors.
Log-Location-Scale	A family of parametric survival models in which the log of survival time follows a location-scale distribution.

Parametric Distribution used in the Study

Weibull Distribution

The Weibull distribution is a widely used probability distribution, originally introduced by Weibull (1951), especially in survival analysis and reliability studies, to model positive and skewed data. Due to its flexibility, it is able to model a wide range of failure behaviors depending on the value of its parameters.

Let T be a random variable following a Weibull distribution with shape parameter $k > 0$ and scale parameter $\lambda > 0$. Then the PDF and CDF of the Weibull distribution are, respectively, given by:

$$f(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1} \exp \left(- \left(\frac{t}{\lambda} \right)^k \right), \quad t > 0$$

$$F(t) = 1 - \exp \left(- \left(\frac{t}{\lambda} \right)^k \right).$$

Moreover, the hazard function is defined as:

$$h(t) = \frac{f(t)}{S(t)} = \frac{k}{\lambda} \left(\frac{t}{\lambda} \right)^{k-1},$$

where $S(t) = \exp \left(- \left(\frac{t}{\lambda} \right)^k \right)$ is the survival function. The behavior of the hazard function depends on the shape parameter k :

- When $k < 1$, the hazard function is decreasing.
- When $k = 1$, it is constant.
- When $k > 1$, it is increasing.

The Log-Normal Distribution

The log-normal distribution is a continuous probability distribution commonly used to model positive and skewed medical data, with its formal statistical foundations established by McAlister (1879). A random variable T

follows a log-normal distribution if its natural logarithm, $Y = \log(T)$, follows a normal distribution with mean μ and variance σ^2 .

For parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, the PDF and CDF of the log-normal distribution are, respectively, expressed as:

$$f(t) = \frac{1}{\sigma t \sqrt{2\pi}} \exp \left[-\frac{(\log t - \mu)^2}{2\sigma^2} \right], \quad t > 0$$

$$F(t) = \Phi \left(\frac{\log t - \mu}{\sigma} \right),$$

where $\Phi(\cdot)$ denotes the CDF of the standard normal distribution. The hazard function is defined as:

$$h(t) = \frac{f(t)}{S(t)} = \frac{\frac{1}{\sigma t \sqrt{2\pi}} \exp \left[-\frac{(\log t - \mu)^2}{2\sigma^2} \right]}{1 - \Phi \left(\frac{\log t - \mu}{\sigma} \right)}.$$

This distribution is suitable for cases where the hazard rate increases to a peak and then decreases over time.

The Log-Logistic Distribution

A random variable T follows a log-logistic distribution if its natural logarithm follow a logistic distribution, which was popularized for survival analysis and mortality modeling (Bennett, 1983). Let $Y = \log(T)$, follow a logistic distribution with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, the PDF and CDF are, respectively, given by:

$$f(t) = \frac{1}{\sigma t} \frac{\exp \left(\frac{\log t - \mu}{\sigma} \right)}{\left[1 + \exp \left(\frac{\log t - \mu}{\sigma} \right) \right]^2}, \quad t > 0$$

$$F(t) = \frac{\exp \left(\frac{\log t - \mu}{\sigma} \right)}{1 + \exp \left(\frac{\log t - \mu}{\sigma} \right)}.$$

The survival function and hazard function are, respectively, written as:

$$S(t) = \frac{1}{1 + \exp \left(\frac{\log t - \mu}{\sigma} \right)},$$

$$h(t) = \frac{1}{\sigma t} \frac{\exp \left(\frac{\log t - \mu}{\sigma} \right)}{1 + \exp \left(\frac{\log t - \mu}{\sigma} \right)}.$$

The Smallest Extreme Value Distribution

The Smallest Extreme Value (SEV) distribution is a continuous probability distribution that arises naturally in the context of extreme value theory, particularly in modeling the minimum of a large number of independent observations. It is closely related to the Weibull distribution, as if a random variable T follows a Weibull distribution, then $Y = \log(T)$ follows a SEV distribution with location parameter $\mu \in \mathbb{R}$ and scale parameter $\sigma > 0$.

The PDF and CDF of the SEV distribution are, respectively, given by:

$$f(y) = \frac{1}{\sigma} \exp\left(\frac{y - \mu}{\sigma}\right) \exp\left(-\exp\left(\frac{y - \mu}{\sigma}\right)\right), \quad y \in \mathbb{R} \quad (1.1)$$

$$F(y) = 1 - \exp\left(-\exp\left(\frac{y - \mu}{\sigma}\right)\right) \quad (1.2)$$

The hazard function is defined as:

$$h(y) = \frac{1}{\sigma} \exp\left(\frac{y - \mu}{\sigma}\right) \quad (1.3)$$

This distribution is particularly important in survival analysis as it serves as the underlying error distribution in the Weibull LLS model.

1.6 Outline of the Research

This research is organized into five main chapters:

Chapter 1 (Introduction): Presents the research problem, highlights its importance, outlines the study objectives, and introduces the key concepts used throughout the research.

Chapter 2 (Literature Review): Reviews relevant statistical and medical studies related to HF survival analysis and parametric modeling.

Chapter 3 (Methodology): Describes the modeling framework adopted in this study, including the LLS family and the AFT regression models (Weibull, Log-normal, and Log-logistic), along with the criteria used for model comparison.

Chapter 4 (Results and Discussion): Presents the exploratory analysis of the HF dataset, the estimation of model parameters, and a comparison of the fitted regression models using selected evaluation criteria.

Chapter 5 (Summary and Future Work): Summarizes the main findings of the study, presents the final conclusions and their practical implications, and provides suggestions for future research and possible extensions of the proposed models.

Chapter 2: Literature Review

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the key studies related to the analysis of HF survival data, with particular emphasis on statistical modeling approaches. Section (2.2) examines research that analyzes HF data within the framework of survival analysis, while section (2.3) focuses on studies that apply the LLS framework through a range of parametric models.

2.2 Heart Failure Data

The HF clinical dataset has been widely used in the literature to investigate patient mortality risk and survival using various modeling techniques. The majority of recent research has relied on deep learning and machine learning methods to increase prediction accuracy. As an instance, Li et al. (2022) applied four machine learning algorithms: XGBoost, logistic regression (LR), random forest (RF), and support vector machine (SVM) to predict the mortality risk for patients with HF in intensive care units. The results demonstrated that the XGBoost model achieved the highest predictive performance among the four models.

Rizinde et al. (2023) predicted risk of hospitalization for HF patients using RF, SVM, LR, K-nearest neighbors, multilayer perceptron, and decision tree. They showed that the RF achieved the best overall performance in predicting the risk of hospitalization for HF patients. Furthermore, Kamio

et al. (2023) utilized SVM, XGBoost, and a multi-layer neural network to predict the mortality of HF patients in the intensive care unit. They demonstrated that SVM and XGBoost have similar moderate performance and outperform the multi-layer neural network. Recently, Hajishah et al. (2025) used several machine learning algorithms, including neural network, RF, LR, SVM, and XGBoost to predict the mortality and hospital readmission for HF patients. The results indicated that the neural network model achieved the highest performance for mortality prediction, while the SVM performed best for readmission prediction.

Although machine learning models are effective in predicting the HF related outcomes, they have certain significant drawbacks. Specifically, they frequently lack interpretability and do not give the survival or hazard function a clear form. Because of this, they provide little information about how the risk of dying changes over time, which is crucial for survival analysis. More comprehensible statistical models, especially semi-parametric ones, have been developed as a result of this restriction. The Cox proportional hazards (CPH) model is among the most popular semi-parametric models in medical research.

For example, Wang et al. (2024) compared the precision of CPH with machine learning models: RF and SVM in predicting the mortality risk of HF patients. They reported that the Cox models outperformed both machine learning models in survival time analysis for HF patients. Mukherjee and Joshif (2024) predicted the risk of death in patients with HF using CPH model, RF, XGBoost, and SVM. They observed that traditional CPH model outperforms rest of the machine learning techniques in predicting the risk of HF death.

While the CPH model is widely used in survival analysis, it suffers from several drawbacks related to semi-parametric models. The CPH model is limited in its ability to describe the underlying risk structure because it does not offer an explicit hazard function for the baseline. In particular, it is unable to accurately depict various hazard shapes, such as increasing, decreasing, or non-monotonic.

Therefore, in order to provide a more thorough description of the survival process, researchers have investigated fully parametric models. As a

result, more parametric models, such as the Weibull, log-normal, and log-logistic distributions are being used to analyze HF data, especially within the framework of the (AFT). For example, Ashine et al. (2024) examined HF data from Ethiopian patients using Bayesian parametric AFT models with shared gamma frailty and Weibull and log-normal distributions. They found that the log-normal AFT model offered the best fit.

Similarly, Montaseri et al. (2025) introduced a multiple kernel learning approach to enhance predictive performance and compared four parametric AFT distributions (Weibull, log-normal, log-logistic, and exponential) on HF data. They reported that the log-normal distribution performed better than the others in terms of AIC and BIC values, and that the incorporation of kernel learning increased predictive stability.

Moreover, Li et al. (2025) evaluated eight parametric AFT distributions on patients with chronic HF and coronary heart disease. They found that the log-normal model consistently produced the best results. These studies support the use of parametric distributions as dependable instruments for parametric survival analysis in the AFT framework in HF research.

Despite these developments, choosing an appropriate parametric model is still difficult since different distributions suggest different hazard behaviors, which could lead to different conclusions. For the purpose of comparing parametric models, a unified framework is thus necessary. Specifically, the LLS family’s use of flexible distributions has the potential to capture a variety of hazard shapes and offer more in-depth understanding of survival dynamics. In order to identify the best model that fits the underlying patterns and offers a more contextualized view of the survival behavior in HF patients, this study compares and applies parametric models Weibull, log-normal and log-logistic to the HF data.

2.3 The Log-Location Scale (LLS) Framework

In the LLS framework, a flexible class of models for lifetime data based on a transformation of the response variable is considered, allowing a wide variety of hazard shapes and distributional features to be modeled. Its effectiveness has been shown in various application domains in several studies.

For example, Silva et al. (2009) introduced an LLS regression model based on the logarithm of an extended Weibull distribution, which is able to capture bathtub-shaped failure rate functions not covered by standard lifetime models, and applied it to engineering reliability data. Louzada and Marchi (2018) further extended the framework by proposing a location regression model based on the complementary exponentiated exponential geometric distribution that enables for increasing, decreasing and bathtub-shaped hazard functions in the presence of censored data in modeling lifetime data.

Moreover, Ramos and Louzada (2019) proposed an LLS regression model and applied it to engineering reliability data and the Stanford heart transplant dataset. They showed that the LLS regression model achieves better performance compared to some existing models. Similarly, Rodrigues et al. (2022) introduced a regression model based on exponentiated odd LLS distributions and their log transformations. They demonstrated how the LLS framework can be used to model migration and demographic data. Overall, these studies show the flexibility and broad applicability of the LLS framework in modeling complex data structures in engineering, medical, and longitudinal studies.

Chapter 3: Methodology

Chapter 3

Methodology

3.1 Introduction

This chapter comprehensively delineates the methodological architecture and statistical frameworks deployed to evaluate HF survival patterns. To establish a robust and mathematically sound analysis, the research transitions systematically from empirical data structure to sophisticated fully parametric modeling.

Specifically, Section (3.2) provides a detailed overview of the clinical dataset and its baseline variables. Section (3.3) establishes the foundational LLS framework and the AFT regression layout. Building upon these core mathematical principles, Sections (3.4), (3.5), and (3.6) unpack the explicit formulations of the Weibull, log-normal, and log-logistic regression models, respectively. To bridge parametric configurations with non-parametric baselines, Section (3.7) incorporates the Kaplan–Meier survival estimation methodology. Finally, Sections (3.8) and (3.9) structuralize the statistical benchmarks for model selection criteria and diagnostic validation procedures, ensuring the overall integrity of our predictive inferences.

3.2 Data Description

Before formulating the mathematical mechanics of our survival frameworks, it is essential to contextualize the empirical evidence under study. Real world clinical data for this investigation were extracted from the med-

ical profiles of 299 HF patients admitted to the Faisalabad Heart Institute and Allied Hospital in Pakistan. This clinical repository, compiled and published by (Chicco and Jurman (2020)), is publicly accessible through the UCI Machine Learning Repository.¹

The dataset incorporates a pivotal time-to-event variable documenting the precise duration in days until patient death or baseline censoring. To pave the way for precise statistical formulation, the clinical, demographic, and biochemical characteristics of the cohort are systematically categorized and summarized in Table 3.1.

Table 3.1: Summary of patient variables with descriptions

Variable	Description	Measurement	Range
Age	Age of the patient	Years	[40, 95]
Anaemia	Presence of anaemia (reduced red blood cells or hemoglobin)	Binary	{0, 1}
High blood pressure	Presence of hypertension	Binary	{0, 1}
Creatinine phosphokinase	Level of CPK enzyme in the blood	$\mu\text{g/L}$	[23, 7861]
Diabetes	Presence of diabetes	Binary	{0, 1}
Ejection fraction	Percentage of blood ejected per contraction	%	[14, 80]
Sex	Sex of the patient	Binary	{0, 1}
Platelets	Platelet count in blood	$10^3/\mu\text{L}$	[25.01, 850.00]
Serum creatinine	Level of creatinine in the blood	mg/dL	[0.50, 9.40]
Serum sodium	Level of sodium in the blood	mEq/L	[114, 148]
Smoking	Smoking status	Binary	{0, 1}
Time	Follow-up time	Days	[4, 285]
Death event	Death during follow-up	Binary	{0, 1}

As seen from Table 3.1, the dataset contains demographic, clinical, and laboratory characteristics along with the event time and censoring indicator.

¹<https://archive.ics.uci.edu/ml/datasets/Heart+failure+clinical+records>

3.3 LLS Framework and AFT Regression

The LLS framework is commonly used to model positive and skewed data, particularly in survival and reliability analysis. In this framework, the response variable is transformed using the natural logarithm, which helps reduce skewness and allows for more flexible modeling of time-to-event data.

Let T denote the survival time. A regression model based on the LLS framework is given by the AFT model:

$$\log(T_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + \sigma \varepsilon_i, i = 1, 2, \dots, n, \quad (3.1)$$

where $\mathbf{x}_i$ is a vector of explanatory variables, $\boldsymbol{\beta}$ is the vector of regression coefficients, $\sigma > 0$ is a scale parameter that control the variability of the survival times, and the error term ε_i follows a specified distribution. In this formulation, survival time is described in terms of a linear framework on a logarithmic scale in conjunction with the explanatory variables. Moreover, the observations are assumed to be independent and follow a specified parametric distribution, and the parameters are estimated using the maximum likelihood estimation (MLE) approach. The regression coefficients $\boldsymbol{\beta}$ can be interpreted in terms of their impact on survival time. In particular, a change in a covariate will result in a multiplicative change in the expected survival time. Consequently, the LLS model is referred to as an AFT model.

This study employs the AFT regression framework to analyze the impact of chosen covariates on survival time within the HF dataset. The model parameters are estimated the maximum likelihood (ML) approach. Moreover, the fitted models are subsequently evaluated using suitable criteria to identify the model that offers the optimal fit to the data.

3.4 Weibull AFT Regression Model

The Weibull distribution is widely used in survival analysis due to its flexibility in modeling different types of hazard behavior. Within the LLS regression framework, the Weibull AFT regression model can be obtained by using the following steps:

$$Y = \log T, \quad (3.2)$$

then,

$$t = e^y, \quad \text{and} \quad \frac{dt}{dy} = e^y. \quad (3.3)$$

Also let,

$$k = \frac{1}{\sigma}, \quad \lambda = \exp(\mu), \quad (3.4)$$

where $T \sim \text{Weibull}(\alpha = \frac{1}{\sigma}, \lambda = \exp(\mu))$.

Based on the transformation theorem, the density function of Y is derived using the relation:

$$f_Y(y) = f_T(y) \left| \frac{dt}{dy} \right|. \quad (3.5)$$

Alternatively, to find the clear form of Y from equation (3.1), one can initiate the process from the standard SEV component $e_i \sim \text{SEV}(0, 1)$, which leads directly to $Y \sim \text{SEV}(\mu, \sigma)$ to complete the solution.

Now, substituting into the model framework, we get:

$$f_Y(y) = \frac{1/\sigma}{e^\mu} \left(\frac{e^y}{e^\mu} \right)^{\frac{1}{\sigma}-1} \exp \left(- \left(\frac{e^y}{e^\mu} \right)^{1/\sigma} \right) e^y. \quad (3.6)$$

Thus,

$$f_Y(y) = \frac{1}{\sigma} \exp \left(\frac{y - \mu}{\sigma} \right) \exp \left\{ - \exp \left(\frac{y - \mu}{\sigma} \right) \right\}. \quad (3.7)$$

Now let,

$$Z = \frac{Y - \mu}{\sigma}. \quad (3.8)$$

Then,

$$Y = \sigma Z + \mu, \quad \text{and} \quad \frac{dy}{dz} = \sigma. \quad (3.9)$$

Substituting into (3.7), the standard smallest extreme value density takes the form:

$$f_Z(z) = \exp(z) \exp\{-\exp(z)\}. \quad (3.10)$$

Now, define

$$z = \frac{\log t - \mu}{\sigma}.$$

Hence

$$f_T(t) = f_Z(z) \left| \frac{dz}{dt} \right|, \quad \frac{dz}{dt} = \frac{1}{\sigma t},$$

Therefore, the density of the Weibull regression takes the form

$$f_T(t) = \frac{1}{\sigma t} \exp(z) \exp\{-\exp(z)\}, \quad t > 0 \quad (3.11)$$

To estimate the model parameters $(\boldsymbol{\beta}, \boldsymbol{\sigma})$, the ML method can be used. Thus, the likelihood function is given by

$$L(\boldsymbol{\beta}, \boldsymbol{\sigma}) = \prod_{i=1}^n [f_T(t_i)]^{\delta_i} [S_T(t_i)]^{1-\delta_i}$$

$$L(\boldsymbol{\beta}, \boldsymbol{\sigma}) = \prod_{i=1}^n \left[\frac{1}{\sigma t_i} \exp(z_i) \exp(-\exp(z_i)) \right]^{\delta_i} [\exp(-\exp(z_i))]^{1-\delta_i},$$

where $S_T(t) = \exp(-\exp(z))$ is the survival function and δ_i is the censoring indicator defined as:

$$\delta_i = \begin{cases} 1 & \text{event (death)} \\ 0 & \text{censored} \end{cases}$$

Thus, the log-likelihood (Logl) is as follows

$$\ell(\boldsymbol{\beta}, \boldsymbol{\sigma}) = \sum_{i=1}^n [\delta_i(-\log \sigma - \log t_i + z_i) - \exp(z_i)] \quad (3.12)$$

Hence, to estimate the model parameters, we differentiate the (Logl) with respect to $\boldsymbol{\beta}$ and $\boldsymbol{\sigma}$ and set the resulting equations equal to zero.

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \frac{\exp(z_i) - \delta_i}{\sigma} x_i$$

$$\frac{\partial \ell}{\partial \sigma} = \sum_{i=1}^n \frac{1}{\sigma} [-\delta_i(1 + z_i) + z_i \exp(z_i)]$$

Thus, the MLEs $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}$ are obtained by equating the above partial derivatives to zero and solving the resulting equations or alternatively solving these equations numerically using the statistical software R.

3.5 Log-Normal AFT Regression Model

The log-normal distribution is a continuous probability distribution commonly used in survival analysis to model positive and skewed data. It is particularly suitable for modeling lifetime data where the hazard rate is non-monotonic and may increase to a peak and then decrease over time.

Within the LLS regression framework, the derivation of the log-normal AFT regression model is expanded step-by-step. Let T follow a log-normal distribution, and define the log-transformation as:

$$Y = \log T. \quad (3.13)$$

By applying the transformation method to the original lifetime variables, we express $t = e^y$ and its derivative $\frac{dt}{dy} = e^y$. Given that $\mu = \mathbf{x}_i^\top \boldsymbol{\beta}$, the random variable Y follows a normal distribution $Y \sim N(\mu, \sigma^2)$. Consequently, the probability density function of Y is expressed as:

$$f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{y - \mu}{\sigma} \right)^2 \right]. \quad (3.14)$$

Now let,

$$z = \frac{y - \mu}{\sigma}. \quad (3.15)$$

Then $y = \sigma z + \mu$, and $\frac{dy}{dz} = \sigma$. Substituting these into the density, the standard normal density takes the form:

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} z^2 \right). \quad (3.16)$$

Now, define z explicitly in terms of the original variables and parameters:

$$z = \frac{\log t - \mu}{\sigma} = \frac{\log t - \mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma}. \quad (3.17)$$

Hence

$$f_T(t) = f_Z(z) \left| \frac{dz}{dt} \right|, \quad \frac{dz}{dt} = \frac{1}{\sigma t}. \quad (3.24)$$

Therefore, the density of the log-normal regression takes the form:

$$f_T(t) = \frac{1}{\sigma t} \phi(z), \quad t > 0, \quad (3.25)$$

where $\phi(\cdot)$ denotes the PDF of the standard normal distribution. Then, the likelihood function can be written as

$$L(\boldsymbol{\beta}, \sigma) = \prod_{i=1}^n \left[\frac{1}{\sigma t_i \sqrt{2\pi}} \exp\left(-\frac{1}{2} z_i^2\right) \right]^{\delta_i} [1 - \Phi(z_i)]^{1-\delta_i},$$

where $1 - \Phi(z_i)$ is the survival function and $\Phi(z_i)$ is the CDF. To estimate the model parameters $(\boldsymbol{\beta}, \sigma)$, the ML method is used. The log-likelihood function is obtained by substituting the explicit expression of z :

$$\ell(\boldsymbol{\beta}, \sigma) = \sum_{i=1}^n \left[\delta_i \left(-\log \sigma - \log t_i - \frac{1}{2} \log(2\pi) - \frac{1}{2} z_i^2 \right) + (1 - \delta_i) \log(1 - \Phi(z_i)) \right] \quad (3.18)$$

The partial derivatives of the log-likelihood with respect to the parameters are:

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \left[\delta_i \frac{z_i}{\sigma} x_i - (1 - \delta_i) \frac{\phi(z_i)}{1 - \Phi(z_i)} \frac{1}{\sigma} x_i \right] \quad (3.19)$$

$$\frac{\partial \ell}{\partial \sigma} = \sum_{i=1}^n \left[\delta_i \frac{z_i^2 - 1}{\sigma} - (1 - \delta_i) \frac{\phi(z_i)}{1 - \Phi(z_i)} \frac{z_i}{\sigma} \right] \quad (3.20)$$

To obtain the MLEs $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}$, setting the above partial derivatives equal to zero and solve the resulting system of equations. Since these equations are nonlinear, they are solved numerically using statistical software such as R.

3.6 Log-Logistic AFT Regression Model

Log-logistic distributions are widely used to represent positive survival data as well as data with skewed distributions. It is particularly suitable for the modeling of lifetime data in which hazard functions are non-monotonic, typically increasing to a peak over time and then decreasing.

Within the LLS regression framework, rather than compressing the initialization, we define the underlying components explicitly. Let T be the survival time following a log-logistic distribution. We apply the log-transformation:

$$Y = \log T, \quad (3.21)$$

which implies $t = e^y$ and $\frac{dt}{dy} = e^y$. Here, the location parameter is structured linearly as $\mu = \mathbf{x}_i^\top \boldsymbol{\beta}$. Under this AFT formulation, the transformed variable Y follows a standard logistic distribution centered at μ with scale parameter σ .

By utilizing the transformation framework, the probability density function of Y is systematically obtained as follows:

$$f_Y(y) = \frac{1}{\sigma} \frac{\exp\left(\frac{y-\mu}{\sigma}\right)}{\left[1 + \exp\left(\frac{y-\mu}{\sigma}\right)\right]^2}. \quad (3.22)$$

Now let,

$$z = \frac{y - \mu}{\sigma}. \quad (3.23)$$

Then $y = \sigma z + \mu$, and $\frac{dy}{dz} = \sigma$. Substituting into the density, the standard logistic density takes the form:

$$f_Z(z) = \frac{\exp(z)}{(1 + \exp(z))^2}. \quad (3.24)$$

Now, define z explicitly in terms of the original variables and parameters:

$$z = \frac{\log t - \mu}{\sigma} = \frac{\log t - \mathbf{x}^\top \boldsymbol{\beta}}{\sigma}. \quad (3.25)$$

Hence

$$f_T(t) = f_Z(z) \left| \frac{dz}{dt} \right|, \quad \frac{dz}{dt} = \frac{1}{\sigma t}. \quad (3.26)$$

Therefore, the density of the log-logistic regression takes the form:

$$f_T(t) = \frac{1}{\sigma t} \frac{\exp(z)}{(1 + \exp(z))^2}, \quad t > 0. \quad (3.27)$$

The likelihood function is given by

$$L(\boldsymbol{\beta}, \sigma) = \prod_{i=1}^n \left[\frac{1}{\sigma t_i} \frac{\exp(z_i)}{(1 + \exp(z_i))^2} \right]^{\delta_i} \left[\frac{1}{1 + \exp(z_i)} \right]^{1-\delta_i}.$$

Hence, log-likelihood function is obtained by

$$\ell(\boldsymbol{\beta}, \sigma) = \sum_{i=1}^n [\delta_i(-\log \sigma - \log t_i + z_i - 2 \log(1 + \exp(z_i))) - (1 - \delta_i) \log(1 + \exp(z_i))], \quad (3.28)$$

where z_i is defined in equation (3.25).

The partial derivatives of the log-likelihood with respect to the parameters are:

$$\frac{\partial \ell}{\partial \boldsymbol{\beta}} = \sum_{i=1}^n \frac{1}{\sigma} \left[\left(\frac{\exp(z_i)}{1 + \exp(z_i)} \right) - \delta_i \right] x_i \quad (3.29)$$

$$\frac{\partial \ell}{\partial \sigma} = \sum_{i=1}^n \frac{1}{\sigma} \left[-\delta_i(1 + z_i) + z_i \frac{\exp(z_i)}{1 + \exp(z_i)} \right] \quad (3.30)$$

To find the MLEs $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}$, setting the above partial derivatives equal to zero and solve the resulting likelihood equations. Because these equations are not closed-form, numerical optimization methods are employed using statistical software such as R.

3.7 Kaplan–Meier Survival Estimation

While the fully parametric designs developed in the previous sections provide detailed insights into survival dynamics, their validity depends on specific distributional assumptions. To evaluate these parametric assumptions without imposing structural constraints, this study incorporates the non-parametric Kaplan–Meier estimation method.

The Kaplan–Meier estimator provides an empirical baseline of survival probability over time (Kaplan and Meier, 1958), calculated as:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i} \right), \quad (3.31)$$

where d_i represents the number of distinct event occurrences (deaths) at time point t_i , and n_i denotes the corresponding number of individuals remaining at risk immediately prior to t_i . This non-parametric baseline allows us to visually inspect survival curves across different categorical groups and provides an empirical standard to validate the fit of our parametric models.

3.8 Model Selection Criteria

In statistical modeling, it is important to assess how well a fitted model describes the observed data and to compare competing models in a systematic way. Model selection and goodness-of-fit measures are used to evaluate the adequacy of different models and to identify the most appropriate one.

3.8.1 Akaike Information Criterion (AIC)

The Akaike Information Criterion (AIC) is defined as:

$$AIC = -2l(\hat{\theta}) + 2k$$

where $l(\hat{\theta})$ is the maximized log-likelihood and k is the number of estimated parameters (Bozdogan, 1987). The AIC evaluates the trade-off between model fit and model complexity by penalizing the inclusion of additional parameters. It is widely used for model comparison, with lower values indicating a better balance between goodness-of-fit and parsimony.

3.8.2 Bayesian Information Criterion (BIC)

The Bayesian Information Criterion (BIC) is given by:

$$BIC = -2l(\hat{\theta}) + k \log(n)$$

BIC applies a heavier penalty for complexity as sample size n increases, favoring simpler models (Schwarz, 1978).

3.8.3 Consistent Akaike Information Criterion (CAIC)

The Consistent Akaike Information Criterion (CAIC) is an extension of the AIC that introduces a stronger penalty for model complexity. It is defined as:

$$CAIC = -2l(\hat{\theta}) + k(\log(n) + 1)$$

CAIC introduces a stronger penalty than AIC, making it more conservative and suitable for large samples to avoid overfitting (Bozdogan, 1987).

For AIC, BIC, and CAIC, smaller values indicate a better-fitting model. The model that consistently provides the most favorable values across these

criteria is selected as the most appropriate for describing the relationship between the covariances and the survival time.

3.8.4 Backward Elimination

Backward elimination is a stepwise variable selection method that begins with the full model containing all candidate predictors, and sequentially removes the least significant variable at each step based on a predefined criterion. In this study, the AIC was used as the selection criterion, where the variable whose removal results in the greatest decrease in AIC is eliminated at each step. This process continues until no further removal improves the model fit, yielding a parsimonious model that balances goodness-of-fit and model complexity.(Hosmer et al., 2008)

3.9 Model Diagnostic and Validation

This section describes the diagnostic and validation processes used to evaluate the performance of the fitted models. Graphical and statistical tools are employed to confirm that the model adequately represents the data. In particular, Cox-Snell residuals are used to measure goodness-of-fit and the variance inflation factor (VIF) is used to measure multicollinearity among predictors.

3.9.1 Cox-Snell Residuals

To assess the goodness-of-fit of the fitted models, Cox-Snell residuals are used as a graphical diagnostic tool. The Cox-Snell residual for the i^{th} observation is defined as:

$$r_i = \hat{H}(t_i),$$

where $\hat{H}(t_i)$ is the estimated cumulative hazard function evaluated at the observed time t_i . If the model fits well, a plot of the cumulative hazard of the residuals against the residuals should approximate a straight line through the origin with a slope of 1 (Cox and Snell, 1968).

3.9.2 Variance Inflation Factor (VIF)

To identify and assess the severity of multicollinearity among the continuous and categorical clinical features in the HF dataset, the VIF is utilized. For a given explanatory variable X_j , the VIF is mathematically expressed as:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad j = 1, 2, \dots, p, \quad (3.32)$$

where R_j^2 represents the multiple coefficient of determination obtained by regressing X_j against all the remaining $p - 1$ independent variables using an ordinary least squares framework (Tan et al., 2019).

The diagnostic evaluation of the calculated VIF_j values relies on standard empirical guidelines:

- $\text{VIF}_j = 1$: Indicates a complete absence of multicollinearity (perfectly orthogonal features).
- $1 < \text{VIF}_j < 5$: Indicates low to moderate collinearity, which typically does not affect the reliability of the estimated regression parameters.
- $\text{VIF}_j \geq 5$: Indicates severe multicollinearity, suggesting that the parameter estimates are poorly determined and corrective actions may be necessary.

In this study, the VIF diagnostic is computed for the final set of generalized clinical predictors following the automated backward elimination procedure to ensure that the statistical inferences drawn from our optimal parametric survival model remain robust and unpolluted by collinear relationships. All computations are conducted numerically within the R statistical environment.

Chapter 4: Results and Discussion

Chapter 4

Results and Discussion

4.1 Introduction

This chapter shows the results obtained from the analysis of the HF dataset using parametric survival models within the LLS framework. To provide a clear and systematic review of the findings, the chapter is divided into five sections. Section (4.2) provides initial insights into the variables and summarizes the key characteristics of the dataset. Section (4.3) identifies the most relevant predictors using backward elimination within the regression framework. Section (4.4) presents the fitted Weibull, Log-Normal, and Log-Logistic models, with addition to their estimated parameters. Section (4.5) determines the best-fitting model using AIC, BIC, and CAIC. Finally, Section (4.6) discusses the selected model, interprets the results, and highlights the key findings.

4.2 Descriptive Analysis

The descriptive analysis provides an initial overview of the HF dataset and summarizes the key characteristics of the variables. The distribution of the binary target variable (death event) across the 299 patients included in this study is presented in Table 4.1. Table 4.1 shows that out of 299 patients, 67.9% survived and 32.1% died by the end of the follow-up period, indicating a moderate class imbalance. In addition, the table shows significant changes in proportions that point to possible clinical risk factors. For example, the deceased patients had a higher percentage of high blood pressure 40.62% than

Table 4.1: Patient characteristics

Category feature	Full sample		Dead patients		Survived patients	
	#	%	#	%	#	%
Anaemia (0: false)	170	56.86%	50	52.08%	120	59.11%
Anaemia (1: true)	129	43.14%	46	47.92%	83	40.89%
High blood pressure (0: false)	194	64.88%	57	59.38%	137	67.49%
High blood pressure (1: true)	105	35.12%	39	40.62%	66	32.51%
Diabetes (0: false)	174	58.19%	56	58.33%	118	58.13%
Diabetes (1: true)	125	41.81%	40	41.67%	85	41.87%
Sex (0: woman)	105	35.12%	34	35.42%	71	34.98%
Sex (1: man)	194	64.88%	62	64.58%	132	65.02%
Smoking (0: false)	203	67.89%	66	68.75%	137	67.49%
Smoking (1: true)	96	32.11%	30	31.25%	66	32.51%

the survived 32.51%. Similarly, anaemia is more common in the deceased group 47.92% than in the survived group 40.89%. In contrast, variables such as sex, smoking status, and diabetes show similar proportions across both groups, indicating weaker associations with mortality at the baseline level. Furthermore, descriptive statistics: the median, the mean, and the standard deviation (SD), of the numerical variables for the full sample and by outcome are presented in Table 4.2.

Table 4.2: Descriptive statistics of numeric features by outcome

Variable	Full sample			Dead			Survived		
	median	mean	SD	median	mean	SD	median	mean	SD
Age	60.00	60.83	11.89	65.00	65.22	13.21	60.00	58.76	10.64
Creatinine phosphokinase	250.00	581.80	970.29	259.00	670.20	1316.58	245.00	540.10	753.80
Ejection fraction	38.00	38.08	11.83	30.00	33.47	12.53	38.00	40.27	10.86
Platelets	262.00	263.36	97.80	258.50	256.38	98.53	263.00	266.66	97.53
Serum creatinine	1.10	1.39	1.03	1.30	1.84	1.47	1.00	1.19	0.65
Serum sodium	137.00	136.60	4.41	135.50	135.40	5.00	137.00	137.20	3.98
Time	115.00	130.30	77.61	44.50	70.89	62.38	172.00	158.30	67.74

The results show noticeable differences between deceased and survived patients. The average age of deceased patients is higher 65.22 years compared to survivors 58.76 years, indicating that older patients are more likely to experience adverse outcomes. In Addition, deceased patients have lower ejection fraction median = 30.00 compared to survivors 38.00, reflecting poorer heart function. Serum creatinine levels are also higher in the deceased group, suggesting worse kidney function. Furthermore, the follow-up time is shorter for deceased patients median = 44.50 days compared to survivors 172.00 days, indicating that adverse events occurred earlier in this group. Overall, these differences highlight the importance of age, cardiac function, and renal function in understanding patient outcomes.

4.3 Variable Selection (Backward Elimination)

An automated backward elimination process was used to determine the most economical and statistically sound subset of predictors from the original pool of clinical features. A parametric Weibull survival distribution framework was used to construct the baseline model, and the selection algorithm was guided by the AIC. Starting from the full model, variables were sequentially removed based on their contribution to the model.

Table 4.3: AIC-guided backward elimination procedure for the Weibull AFT survival model.

Step	Variable Removed	AIC	Δ AIC
0	Full model (all predictors)	1282.76	—
1	Smoking	1280.90	−1.86
2	Platelet count	1279.21	−1.69
3	Diabetes	1277.60	−1.61
4	Blood pressure (hypertension)	1276.10	−1.50
5	Anaemia status	1274.62	−1.48

Note: Δ AIC = change in AIC relative to the previous step. Variables are listed in the order of removal. The final model at Step 5 achieves the minimum AIC of 1274.62, retaining six predictors: age, ejection fraction, serum creatinine, serum sodium, sex, and blood pressure.

The initial full model contained nine active primary clinical predictors. After completing the backward elimination process, the final model retained six predictors: age, ejection fraction, serum creatinine, serum sodium, sex, and blood pressure. Figure 4.1 shows a scatter plot of serum creatinine versus ejection fraction, color-coded by patient mortality status. The plot shows

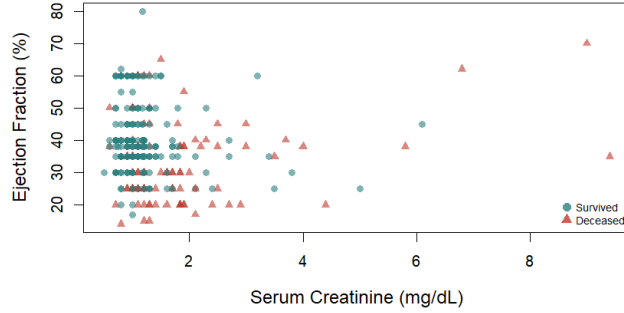


Figure 4.1: Scatter Plot of Serum Creatinine vs. Ejection Fraction by Survival Status

that patients who died during the follow-up period are highly concentrated in the region that represents both low ejection fraction and high serum creatinine values. This pattern indicates that patients with reduced cardiac output and impaired renal function are more likely to experience adverse outcomes. The figure provides a visual indication of the importance of these variables in the model.

4.4 Parametric Survival Models

Three different parametric AFT survival models the Weibull, log-normal, and log-logistic models were fitted using the six selected variables obtained from the backward elimination process. Each model was fitted under the same conditions to allow a fair comparison of their performance. Four information criteria and likelihood measures were used to compare the models and identify the best-fitting model: logL, AIC, BIC, and CAIC.

Table 4.4: Parametric Survival Model Comparison Statistics

Model	Logl	AIC	BIC	CAIC
Weibull	-630.812	1277.624	1307.327	1315.237
Log-Normal	-633.279	1282.557	1312.161	1320.161
Log-Logistic	-632.179	1280.359	1309.962	1317.962

The results in Table 4.4 show that the Weibull model outperforms both the log-logistic and log-normal models. as it achieves the lowest AIC, BIC, and CAIC, along with the highest logL value among the competing models. Based on these results, the Weibull model is selected as the best-fitting model for this dataset, as it provides the best balance between model fit and model complexity.

4.4.1 Kaplan–Meier Survival Curves

Figures 4.2–4.4 present the Kaplan–Meier survival curves for different clinical variables.

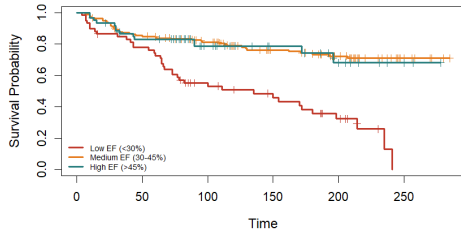


Figure 4.2: Kaplan-Meier Survival Curves by Ejection Fraction

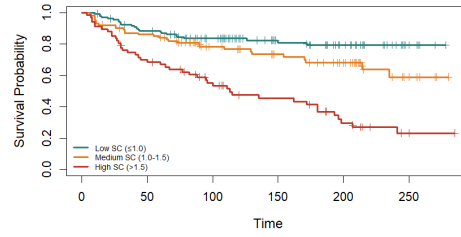


Figure 4.3: Kaplan-Meier Survival Curves by Serum Creatinine

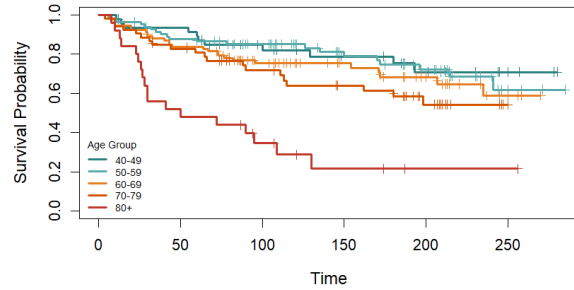


Figure 4.4: Kaplan-Meier Survival Curves by Age Group

From these figures, it is clear that patients with lower ejection fraction exhibit a sharper decline in survival probability compared to those with higher values, reflecting poorer heart function. Similarly, higher serum creatinine levels are associated with reduced survival probabilities, indicating worse outcomes and impaired kidney function. In addition, older patients tend to have lower survival probabilities compared to younger groups, showing a clear effect of age on survival. Overall, these patterns highlight noticeable differences between groups and suggest that these variables play an important role in explaining patient outcomes.

4.5 Model Refinement and Validation

The selected Weibull model was further refined by evaluating the significance of its six variables. Each variable's individual significance was assessed using its partial P -value and standard error baseline within the final model estimation to identify a more refined subset of predictors. Table 4.5 presents the Weibull regression coefficients, along with the standard error and the P -value for the six variables.

Table 4.5: Weibull Regression Coefficients, Standard Error, and the P -value for the Six Variables

Variable	Coefficient (β)	Standard Error	P -value
Blood Pressure	-0.35266	0.32065	0.27131
Age	-0.04743	0.00971	0.00103
Ejection Fraction	0.05043	0.01132	0.00100
Serum Creatinine	-0.31606	0.07277	0.00141
Serum Sodium	0.04570	0.02891	0.11394
Sex	-0.10606	0.08708	0.22323

From Table 4.5, three of the six variables (blood pressure, serum sodium, and sex) were removed during this refinement stage because they did not meet the significance threshold ($\alpha = 0.05$). With a highly significant P -value of 0.001 for each, age, ejection fraction, and serum creatinine remain the key variables in the final model. These variables show strong effects and are considered the most important predictors of survival time in the dataset. Figures 4.5 and 4.6 provide diagnostic checks for the fitted Weibull model. As seen from these figures, the fitted cumulative hazard curve closely follows

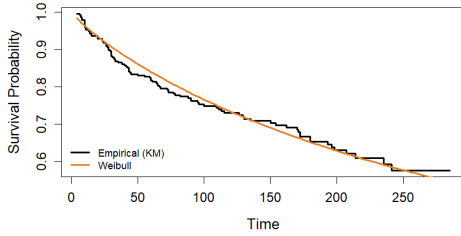


Figure 4.5: Weibull Fitted vs. Empirical Cumulative Hazard

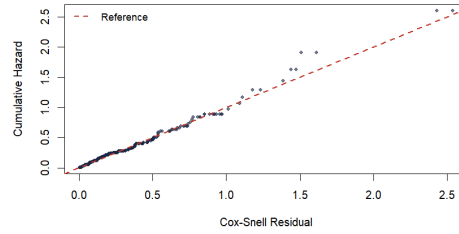


Figure 4.6: Cox-Snell Residual Plot for Model Fit Validation

the empirical cumulative hazard, indicating a good model fit. In addition, the Cox-Snell residual plot shows that the points closely follow the reference line, suggesting that the model provides an adequate representation of the data and fits the survival pattern well.

Finally, a diagnostic test for multicollinearity was carried out by computing the VIF for the six variables before refinement. The VIF values are listed in Table 4.6. All VIF values are ranged from 1.012 to 1.065, which are

Table 4.6: Variance Inflation Factors (VIF) for the Generalized Predictors

Variable	Variance Inflation Factor (VIF)
Age	1.043
Anemia	1.013
Ejection Fraction	1.065
Serum Creatinine	1.012
Serum Sodium	1.041
Sex	1.046

significantly below the conservative threshold of 5.0, confirming that multicollinearity is not a concern among the predictors. This means that the variables do not show strong linear dependence and can be used together in the model without affecting stability.

4.6 Final Model and Discussion

The final optimized model is based on the three most important clinical variables and follows a Weibull parametric regression framework. The computed regression coefficients (β) provide useful insights into the dynamics of patient survival. Age has a negative coefficient ($\beta = -0.04743$, P -value = 0.001), indicating an inverse relationship with survival time. This means that increasing age is associated with a decrease in survival duration. Serum creatinine also has a negative coefficient ($\beta = -0.31606$, P -value = 0.001). Elevated serum creatinine levels indicate kidney distress and are associated with worse patient outcomes and reduced survival time. Moreover, ejection fraction shows a positive coefficient ($\beta = 0.05043$, P -value = 0.001), indicating a protective effect. Higher ejection fraction values reflect better cardiac function and are associated with longer survival time.

In conclusion, the model shifts from a larger set of variables to a more refined structure based on three key predictors. The final model highlights

the importance of age, cardiac function, and renal function in explaining survival outcomes, while maintaining a good balance between model simplicity and predictive performance.

Chapter 5: Summary and Future Work

Chapter 5

Summary and Future Work

5.1 Introduction

The main findings of the research are presented in this chapter, which also discusses the study's implications and suggests possible directions for future research in the context of heart failure survival analysis. Section (5.2) presents the summary of findings, highlighting the main results obtained from the analysis of the HF dataset. Section (5.3) provides the main conclusions, and section (5.4) outlines possible directions for future work.

5.2 Summary

In this study, a parametric AFT model was used to examine the prognostic factors affecting survival time in patients with HF. To improve the model parsimony, an automated backward elimination method has been adopted to reduce the initial clinical profile of 12 features, based on the AIC. The method reduced the covariate to six variables; age, ejection fraction, serum creatinine, serum sodium, sex and blood pressure.

The six predictors were tested with three parametric distributions: Weibull, log-normal and log-logistic models. The Weibull distribution was found to be the most suitable model for this patient cohort by the relative information criterion and likelihood metrics: logL, AIC, BIC and CAIC.

A later modification based on significance of parameters reduced the

model to three important variables: age, serum creatinine and ejection fraction. The estimated coefficients showed a negative effect of age and serum creatinine and a favorable effect of ejection fraction on survival time.

Model performance was evaluated using fitted and empirical hazard and Cox-Snell residuals analysis. The results showed that the fitted Weibull model fits well with the empirical survival pattern, indicating a good overall fit.

5.3 Conclusion

The results of this study showed that the Weibull parametric model was adequate to model the survival time of patients with HF. The final model showed the importance of three main clinical variables: age, serum creatinine, and ejection fraction. Age was found to have a negative effect on survival time, in particular older patients were more prone to have worse results. Serum creatinine had a negative effect suggesting that reduced renal function is associated with reduced survival. In contrast, ejection fraction had a positive effect, indicating that the better the heart works, the longer a person lives.

In conclusion, the present findings suggest that age, cardiac function and renal function are important factors to be considered for determining the survival of patients. It offers a simple and effective representation of the data, achieving an appropriate trade-off between accuracy and model complexity.

5.4 Future Work

In future studies, more advanced modeling techniques could be applied to improve predictive accuracy and reveal complex survival patterns. An example of this is the combination of machine learning techniques and the parametric Weibull framework. The AFT model can be combined with RF or neural networks to improve the accuracy of the forecasts while keeping the interpretability of the parametric framework.

Another important extension is to specify a multi-state survival model.

The model can be extended to include intermediate health states such as hospitalization or disease progression rather than the simple transition between survival and death. This offers a clearer view of how patient conditions evolve and a better understanding of the natural history of HF over time.

These expansions could contribute to improving the understanding of survival dynamics and improving the practical applicability of survival models in clinical settings.

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الملخص

يُعدّ فشل القلب من الأسباب الرئيسية للوفاة على مستوى العالم، ويتميز بأنماط بقاء متباينة بشكل كبير. في هذه الدراسة، طُبِّقت تحليلات البقاء البارامترية ضمن إطار نماذج موقعمقياس اللوغاريتم على مجموعة بيانات سريرية لمرضى فشل القلب، بهدف تفسير ديناميكيات الزمن حتى الحدث وتحديد التنبؤات السريرية الرئيسية. أسفر إجراء الاختيار التراجعي الآلي المُحسّن بمعيار معلومات أكايكي عن تقليص المتغيرات السريرية الأولية إلى مجموعة أصغر من المتغيرات المعمة، شملت: العمر، والكسر القذفي، والكرياتينين المصلي، والصوديوم المصلي، والجنس، وضغط الدم. وقد كشفت المقارنة بين النماذج وفق معايير المعلومات أن توزيع ويبول هو الإطار الأمثل. وبعد استبعاد المتغيرات غير الدالة إحصائياً، تبيّن أن العمر والكسر القذفي والكرياتينين المصلي هي المتغيرات الجوهرية المحركة لتشخيص المريض. وقد أشارت النتائج إلى أن التقدم في العمر وارتفاع مستوى الكرياتينين المصلي يُقلّلان من زمن البقاء، في حين أبدى الكسر القذفي المرتفع أثراً وقائياً. وقد جرى التحقق من جودة ملائمة النموذج النهائي باستخدام منحنيات كابلان ماير ومنحنى الخطر التراكمي المُقدّر، مما أثبت حسن الملاءمة العامة للنموذج.

الكلمات المفتاحية: زمن الفشل المتسارع، فشل القلب، موقعمقياس اللوغاريتم، توزيع ويبول، النمذجة البارامترية

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

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إعداد:

جود باسم البلاجي - ٠٩١٢١٣٢
ميرال هاشم الجبني - ١٢٢٧١٣٢

إشراف:

د. دوله السلمي

كلية العلوم - قسم الإحصاء
جامعة الملك عبدالعزيز
جدة، المملكة العربية السعودية

يونيو ٢٠٢٦



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ميرال هاشم الحجهي - ١٢٢٧١٣٢

إشراف:

د. دولة السلمي

كلية العلوم - قسم الإحصاء
جامعة الملك عبدالعزيز
جدة، المملكة العربية السعودية

يونيو ٢٠٢٦